



When to use GRU over LSTM?

Asked 8 years, 1 month ago Modified 2 years, 2 months ago Viewed 186k times

The key difference between a GRU and an LSTM is that a GRU has two gates (*reset* and *update* gates) whereas an LSTM has three gates (namely *input*, *output* and *forget* gates).

183 Why do we make use of GRU when we clearly have more control on the network through the LSTM model (as we have three gates)? In which scenario GRU is preferred over LSTM?

[neural-network](#) [deep-learning](#) [lstm](#) [gru](#)



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edited Dec 8, 2019 at 10:16

Djib2011
8,028 5 28 38

asked Oct 17, 2016 at 11:47

Sayali Sonawane
2,091 3 13 13

4 A GRU is slightly less complex but is approximately as good as an LSTM performance-wise. An implementation in TensorFlow is found here: [data-blogger.com/2017/08/27/gru-implementation-tensorflow](#). – [www.data-blogger.com](#) Aug 27, 2017 at 12:28

GRUs are generally used when you do have long sequence training samples and you want a quick and decent accuracy and maybe in cases where infrastructure is an issue. LSTMs are preferred when sequence lengths are more and some good context is there. LSTMs when trained with more data give you better results than GRUs. – [Subir Verma](#) Oct 18, 2020 at 8:03

3 @SubirVerma, doesn't "long sequence training samples" and "sequence lengths are more" mean the same thing? – [Naveen Reddy Marthala](#) Jun 26, 2022 at 11:30

6 Answers

Sorted by: Highest score (default)

GRUs and LSTMs utilize different approaches toward gating information to prevent the vanishing gradient problem. Here are the main points comparing the two:

- The GRU unit controls the flow of information like the LSTM unit, but without having to use a **memory unit**. It just exposes the full hidden content without any control.
- GRUs are relatively new, and in my experience, their performance is on par with LSTMs, but computationally **more efficient** (as pointed out, they have a less complex structure). For that reason, we are seeing it being used more and more.

For a detailed description, you can explore this research paper on [Arxiv](#). The paper explains all this brilliantly.

You can also explore these blogs for a better idea:

- [WildML](#)
- [Colah - Github](#)

Hope that helps!

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edited Aug 26, 2022 at 22:08
user133478

answered Oct 17, 2016 at 13:33
 Abhishek
1,989 3 15 21

1 In addition to your answer there is a nice paper evaluating the performance between GRU and LSTM and their various permutations: ["An empirical exploration of recurrent network architectures"](#) by Google – [tastyminerals](#) Jun 10, 2017 at 18:11 ✓

@abhishek I found your answer is a bit counter intuitive - how GRU "without having memory unit" can have a better performance - perhaps I should read the other papers – [user702846](#) Jan 22, 2020 at 13:45

*To complement already great answers above.

- From my experience, **GRUs train faster** and perform better than LSTMs on **less training data** if you are doing language modeling (not sure about other tasks).
- GRUs are simpler** and thus easier to modify, for example adding new gates in case of additional input to the network. It's just less code in general.
- LSTMs** should in theory **remember longer sequences** than GRUs and outperform them in tasks requiring modeling long-distance relations.

*Some additional papers that analyze GRUs and LSTMs.

- "Neural GPUs Learn Algorithms" (Łukasz Kaiser, Ilya Sutskever, 2015) <https://arxiv.org/abs/1511.08228>
- "Comparative Study of CNN and RNN for Natural Language Processing" (Wenpeng Yin et al. 2017) <https://arxiv.org/abs/1702.01923>

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edited Sep 27, 2017 at 13:06
 Neil Slater
29.2k 5 81 100

answered Jun 10, 2017 at 18:22
 tastyminerals
2,157 3 18 20

Good balance of specific precise details while yet concise. – [WestCoastProjects](#) Feb 20, 2021 at 16:57

FULL GRU Unit

$$\tilde{c}_t = \tanh(W_c[G_r * c_{t-1}, x_t] + b_c)$$

$$G_u = \sigma(W_u[c_{t-1}, x_t] + b_u)$$

$$G_r = \sigma(W_r[c_{t-1}, x_t] + b_r)$$

$$c_t = G_u * \tilde{c}_t + (1 - G_u) * c_{t-1}$$

$$a_t = c_t$$

LSTM Unit

$$\tilde{c}_t = \tanh(W_c[a_{t-1}, x_t] + b_c)$$

$$G_u = \sigma(W_u[a_{t-1}, x_t] + b_u)$$

$$G_f = \sigma(W_f[a_{t-1}, x_t] + b_f)$$

$$G_o = \sigma(W_o[a_{t-1}, x_t] + b_o)$$

$$c_t = G_u * \tilde{c}_t + G_f * c_{t-1}$$

$$a_t = G_o * \tanh(c_t)$$

As can be seen from the equations LSTMs have a separate update gate and forget gate. This clearly makes LSTMs more sophisticated but at the same time more complex as well. There is no simple way to decide which to use for your particular use case. You always have to do trial and error to test the performance. However, because GRU is simpler than LSTM, GRUs will take much less time to train and are more efficient.

Credits:Andrew Ng

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edited Oct 2, 2019 at 5:52
 Martin Alcubierre
193 7

answered Aug 8, 2018 at 3:48
 balboa
351 2 2

On the next slide after the hand written ones, the last equation is different: $a^{<t>} = \Gamma_o \odot \tanh(\tilde{c}^{<t>})$. This formula is confirmed correct [here](#). – Tom Hale Mar 25, 2019 at 8:04



16



This answer actually lies on the dataset and the use case. It's hard to tell definitively which is better.

- GRU exposes the complete memory unlike LSTM, so applications which that acts as advantage might be helpful. Also, adding onto why to use GRU - it is computationally easier than LSTM since it has only 2 gates and if it's performance is on par with LSTM, then why not?
- This [paper](#) demonstrates excellently with graphs the superiority of gated networks over a simple RNN but clearly mentions that it cannot conclude which of the either are better. So, if you are confused as to which to use as your model, I'd suggest you to train both and then get the better of them.

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edited Oct 17, 2016 at 13:35

answered Oct 17, 2016 at 12:13
 Hima Varsha
2,356 15 34



-2



GRU is better than LSTM as it is easy to modify and doesn't need memory units, therefore, faster to train than LSTM and give as per performance.

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answered Dec 5, 2017 at 4:12
 Vivek Khetan
367 1 7

34 please support the performance claim with fair references – Kari May 21, 2018 at 3:57



-2



Actually, the key difference comes out to be more than that: Long-short term (LSTM) perceptrons are made up using the momentum and gradient descent algorithms. When you reconcile LSTM perceptrons with their recursive counterpart RNNs, you come up with GRU which is really just a generalized recurrent unit or Gradient Recurrent Unit (depending on the context) that more closely integrates the momentum and gradient descent algorithms. Were I you, I'd do more research on AdamOptimizers.

GRU is an outdated concept by the way. However, I can understand you researching it if you want moderate-advanced in-depth knowledge of TF.

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edited Oct 26, 2018 at 21:40

answered Oct 26, 2018 at 21:04
 Andre Patterson
15 3

24 I'm curious. Could you explain why GRU is an outdated concept? – random_user Nov 16, 2018 at 18:01

3 If you make a claim that the primary topic of the question is "an outdated concept" you *really* should back that up with references (as mentioned in prior comment) – WestCoastProjects Feb 20, 2021 at 16:59



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